



STREETLIGHT

INSIGHT

StreetLight AADT 2023
Methodology and Validation
White Paper
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Introduction

Annual Average Daily Traffic (AADT) is the industry's fundamental metric for analyzing infrastructure projects, measuring congestion, estimating road safety, and enabling empirical measurement for approving highway funds. Traditionally, AADT counts are collected via permanent or temporary physical sensors, limited to a fraction of U.S. roadways. The StreetLight Data, Inc. ("StreetLight") InSight® platform delivers AADT for virtually any roadway on-demand, directly through the computer.

Modeling AADT counts from regional travel demand models often leads to inaccuracies, as does temporary count expansion. Additionally, temporary counts are cumbersome and expensive. StreetLight develops an AADT estimate each year for urban and rural road segments across the U.S. that performs as well as or better than industry-standard accuracy targets and can be used by industry practitioners for traffic monitoring and reporting, traffic-impact studies, infrastructure prioritization, and more. Year after year, StreetLight's data science team continuously optimizes the AADT metric to bring the most comprehensive estimates to agencies across the U.S.

What's New in AADT 2023

AADT 2023 includes metrics powered by Aggregated GPS Data. This includes a unique blend of device navigation apps, traditional mobile data apps, and in-vehicle navigation apps. Aggregated GPS Data provides some advantages in that it has a highly representative sample with strong penetration rates across road classes or types and regions. Penetration rates are measured as sample trip counts divided by ground-truth vehicle volumes collected at permanent counter locations. In an evaluation of penetration rates across U.S. roads in 2023, we found that, on average, our Aggregated GPS Data source captured 27% of vehicles on the road. This highly representative sample contributes to strong metrics quality, especially on rural and lower-volume roads.

AADT 2023 is built directly on the results of the AGPS monthly volume model. For details on the data sources and model design used in the AGPS Volume model, please see [the white paper](#). Estimated AADT 2023 is equal to the average of Monthly Average Daily Traffic (MADT) estimates for all 12 months in 2023, weighted by the number of days in the month.

This paper is an update to the AADT 2023 v1 metric, originally released in April 2024. AADT 2023 v2 benefits from the addition of many more permanent counters included in the training set along with improvements to the algorithm refining the model inputs, leading to enhanced performance.

StreetLight provides AADT estimates from 2019 through 2023. These estimates can be compared

across time in order to assess annual change on a roadway or applied to use cases like before-and-after studies.

Ground-Truth Data

To train and validate the AADT 2023 metric, we used vehicle volume counts from 7,686 unique permanent counter locations across all states in the contiguous U.S. This is an improvement over our v1 model, which included data from 22 states, as not all states had released AADT counts for 2023 at the point of model development.

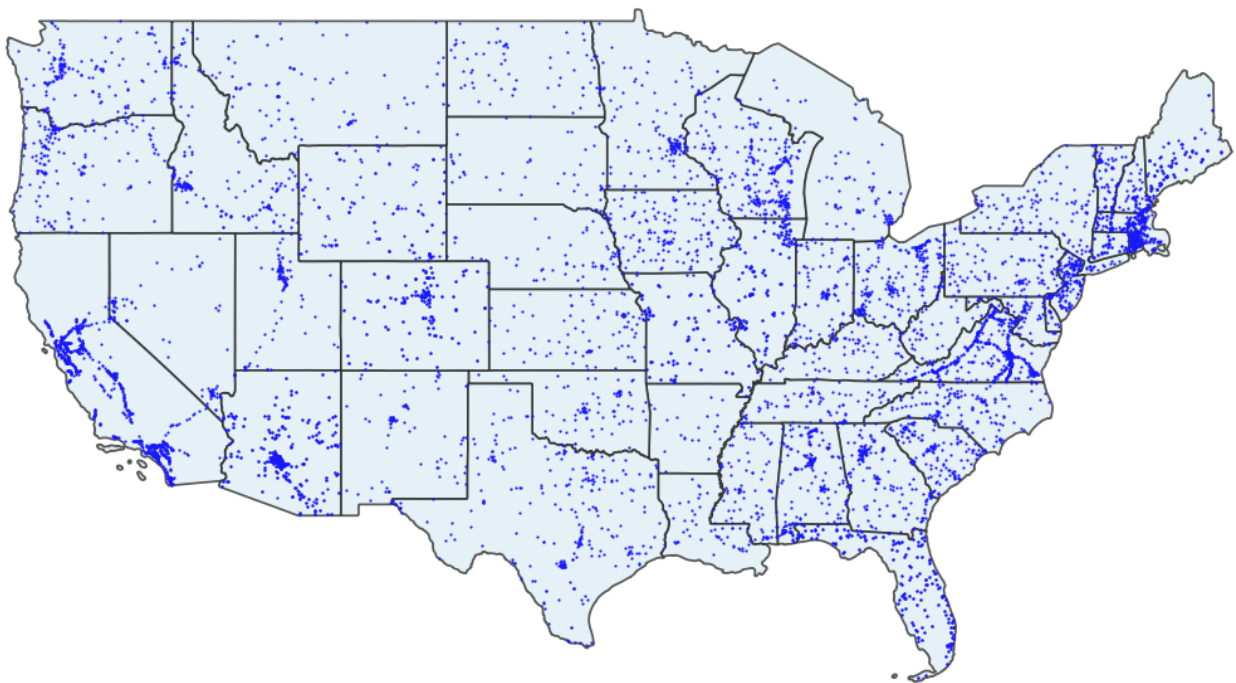


Figure 1: Location of permanent counters used to train and validate the StreetLight AADT 2023 U.S. metric.

Evaluating Accuracy of StreetLight's AADT 2023

The following sections describe accuracy definitions and present validation results for AADT 2023. To measure the accuracy of our AADT 2023 metric, we compare AADT estimates to the ground truth data from permanent counters. Then, we evaluate the accuracy of those estimates against the industry-standard benchmark, short-term count expansion.

Defining Error Metrics

To evaluate model accuracy, we consider the following five error metrics, each of which illuminates a different aspect of model performance. Metrics are reported by road size, since

errors (measured as a percentage of permanent counter AADT) naturally increase along smaller roads. The selected error metrics align with those that are most widely utilized in the transportation industry.

- **Mean absolute percentage error (MAPE)** is a standard measure of error for predictive machine-learning models. It describes errors well on small roads and treats errors of all sizes equally.
- **Normalized root mean square error (NRMSE)**, another standard error metric in machine learning, penalizes large errors more, making it more sensitive to the accuracy of AADT estimation on high-volume roads.
- **68th percentile absolute error** represents one standard deviation from the mean within a standard bell curve and thus describes the typical error across the road segments. The 68th percentile absolute error can be interpreted as the expected typical error.
- **95th percentile absolute error** represents the upper limit of the expected absolute error.
- **Median percentage error** is a measure of bias in our model, which also indicates how accurate an aggregate value (such as the sum of volume on all roads) will be. Values close to zero suggest that our model has low bias, while positive values would indicate more overestimation and negative values indicate more underestimation.

Sources of Error

StreetLight AADT 2023 is a sample expansion, and sampling bias over time and space is the most common cause of error. Though we account for much of the variation in the sample with our AGPS Volume Model that underpins AADT, we expect higher error levels in regions or on road types or classes where we do not have permanent counters.

Additionally, StreetLight relies on an algorithm that maps road segments from the segment library of our Aggregated GPS data provider to the StreetLight InSight® Zone Library, derived from OpenStreetMap (OSM). Significant differences in the two segment maps, while rare, are another source of error. Segment mapping issues are most common where many roads intersect, such as busy highway interchanges, or on highly circuitous segments in rural and residential areas.

Short-Term Count Expansion as an Accuracy Target

We compare StreetLight's AADT model with alternative industry-standard estimation approaches. As a benchmark, we consider one widely used approach: short-term count expansion. In this approach, a two-day temporary count is collected from a road segment and then expanded to estimate AADT using scaling factors from nearby permanent counters. Table 1 shows expected errors associated with a same-year, short-term count expansion for 10

categories of roads, as estimated by the FHWA.¹ Our aim was for StreetLight's AADT accuracy to exceed or come close to the existing short-term count expansion accuracy targets.

- The benchmarks shown in Table 1 are very optimistic because same-year, short-term counts are often unavailable. In many cases, expansion factors are developed using nearby counter data from a previous year, and those are prone to higher error. As many as half of the roads with less than 55,000 AADT lack same-year permanent counter data, according to an unpublished analysis by Minnesota DOT.²
- For roads with less than 500 AADT, there is no accepted target accuracy due to lack of data.

Road Size (AADT)	Median Error (Bias) (%)	68 Percentile Absolute Error (%)	95 Percentile Absolute Error (%)	MAPE (%)	NRMSE (%)
A: 0 – 499	Expected errors not available due to lack of data.				
B: 500 – 1,999	0.0	11.7	26.9	10.0	12.9
C: 2,000 – 4,999	2.2	10.7	33.5	10.4	17.2
D: 5,000 – 9,999	3.1	9.6	28.1	9.2	13.9
E: 10,000 – 19,999	1.2	9.2	27.8	8.9	12.9
F: 20,000 – 34,999	0.8	8.3	24.3	8.1	13.2
G: 35,000 – 54,999	0.4	8.3	19.3	7.2	9.7
H: 55,000 – 84,999	-0.2	6.0	14.4	5.2	7.2
I: 85,000 – 124,999	0.0	4.8	14.7	4.6	6.7
J: 125,000+	3.0	7.0	17.6	6.2	9.9

Table 1: Errors expected from short-term count expansion estimation of AADT serve as our target accuracy. Errors are shown across 10 categories of road size for a same-year, 48-hour count. Data from Krile et al. (2015).¹

In cases where short-term count expansion is not available, the errors from alternative methods would be even higher. Table 2 displays expected error statistics for four categories of roads: very low, low, medium, and high traffic volume. In this table, we include estimated accuracy targets

¹ Krile, R., Todt, F., Schroeder, J. (2015). Assessing Roadway Traffic Count Duration and Frequency Impacts on Annual Average Daily Traffic Estimation (FHWA-PL-16-008). United States. Federal Highway Administration.

² Minnesota Department of Transportation (2005-2019). [HPMS and Historical AADT Data] [Unpublished Raw Data].

for the short-term count expansion method, as well as the more common situation with no counts for the same calendar year.³

- Accuracy estimates are not available for very low-volume roads (AADT < 500), and 68th percentile and 95th percentile absolute percentage error estimates are not available for the “no count” method for any road size.

Road Size (AADT)	Method	Median Bias (%)	MAPE (%)	NRMSE (%)
0 – 499 (very low)	Short-term count expansion	Expected errors not available due to lack of data.		
	Typical “no count” estimates			
500 – 4,999 (low)	Short-term count expansion	-0.1	10.2	18.0
	Typical “no count” estimates	Not available	50	65
5,000 – 54,999 (medium)	Short-term count expansion	1.1	8.6	14.2
	Typical “no count” estimates	2	18	27
55,000± (high)	Short-term count expansion	1.4	5.3	9.5
	Typical “no count” estimates	1.5	20	12

Table 2: Estimated accuracy of short-term count expansion as represented by Krile et al. (2015) and typical situations with no counts from the same calendar year.

Cross-Validation Results

We evaluate accuracy using cross-validation, a technique in which a subset of the ground-truth data is held out while the model is trained on the remaining data. Then the error metrics are calculated using the holdout group. The results represent the expected error at a location on which the model has not been trained. In this section, we report the results of k-fold cross-validation, where “k” equals 10.

Predictive Power

StreetLight 2023 AADT estimates strongly correlate with ground-truth AADT, as shown in Figure 2. The best-fit line’s r-squared is 0.98.

³ Hallenbeck, M., Schewel, L., Co, S., Wergin, J. (2021). “Guidelines for Obtaining AADT Estimates from Non-Traditional Sources,” Non-Traditional Methods to Obtain Annual Average Daily Traffic (AADT) (FHWA-PL-21-031). United States. Federal Highway Administration.

Correlation between Counter AADT and Estimated AADT

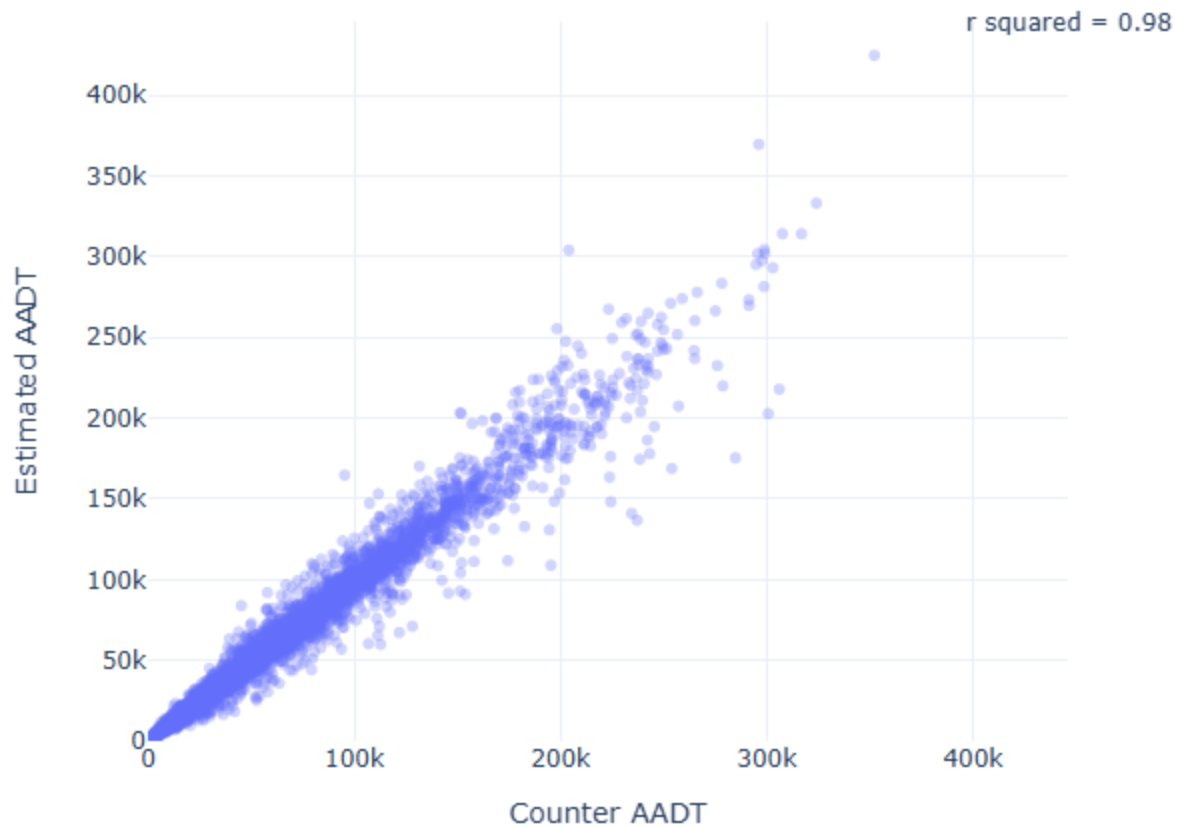


Figure 2: Estimated AADT 2023 values correlate very closely with those reported by permanent counters.

Comparison to Benchmarks

Estimation accuracy for StreetLight's AADT 2023 metric is shown in Table 3 below. The cell shading indicates model accuracy compared to expected short-term count expansion accuracy from Table 1.

- The darker green shading highlights where StreetLight's AADT metric is less than or within one percentage point of the benchmark.
- The light green shading indicates where it is within five percentage points. For example, for road size C, StreetLight's AADT MAPE is 12.1%, while the expected short-term count expansion MAPE is 10.4%. Since 12.1% is within five percentage points of 10.4%, the cell is shaded light green.
- Since median bias can be either positive or negative, we compare the absolute values of the errors.

Road Size (AADT)	Number of Counters	Median Bias (%)	68th Percentile Absolute Error (%)	95th Percentile Absolute Error (%)	MAPE (%)	NRMSE (%)
A: 0 – 499*	321	3.5	22.9	51.8	19.3	24.9
B: 500 – 1,999	1446	0.5	17.3	38.2	14.5	19.8
C: 2,000 – 4,999	1975	-0.7	14.5	32.7	12.2	16.6
D: 5,000 – 9,999	1984	-1.5	12.3	27.8	10.3	13.7
E: 10,000 – 19,999	2279	-1	10.5	24.7	9	12
F: 20,000 – 34,999	1904	-0.8	9.6	23.3	8.3	11.4
G: 35,000 – 54,999	1353	-1	8.9	23.8	8.1	11.9
H: 55,000 – 84,999	1343	-2.2	8.5	21.9	7.6	10.7
I: 85,000 – 124,999	772	-1.9	8.7	22.2	7.9	10.9
J: 125,000 +	691	-2.3	8.9	22.3	7.7	12

Table 3: Cross-validation results for StreetLight AADT 2023 by road size, shaded to compare with short-term count expansion errors.

*No short-term count targets are available for roads of 0–499 AADT, so this row is unshaded.

The cross-validation results reveal that, overall, StreetLight’s AADT model achieves an accuracy similar to that expected from same-year temporary count expansion.

- Error metrics for most road sizes are within 5 percentage points of the errors associated with same-year temporary counter expansion.
- Bias is very low – within 5 percentage points – on all road sizes. This means that, on average, we do not systematically over or underestimate AADT to a significant degree. Bias is typically negative, which means the StreetLight AADT metric errs on the side of very slight underestimation.
- Expected short-term count errors are not available for roads with less than 500 AADT, so road size A cannot be compared.

As previously noted, same-year temporary counts are often unavailable. Compared to a typical “no count” situation, StreetLight AADT performs very well. In Figure 3, we compare the MAPE of StreetLight AADT estimates to both short-term count expansion and “no count” estimation. Though StreetLight MAPE is slightly higher than that of short-term count expansion, both are much better than the MAPE associated with “no count” estimation.

While we do not have benchmarks for road size A (0–499), our error levels on these roads are within expectations. We expect very low volume roads to have the highest error levels due to limited sample size. Though error is in fact highest on these roads, it is still lower than the “no count” error level on even larger roads.

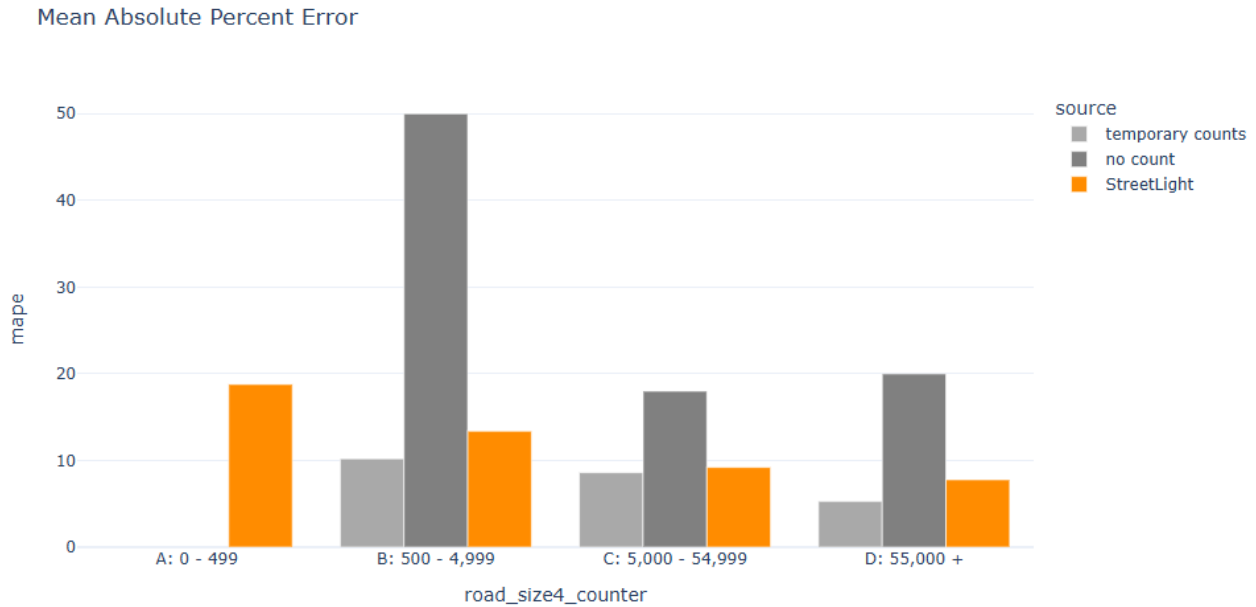


Figure 3: StreetLight mean absolute percent error compared to temporary count expansion and “no count” volume estimation benchmarks.

About StreetLight Data

StreetLight Data, Inc. (“StreetLight”) pioneered the use of Big Data analytics to shed light on how people, goods, and services move, empowering smarter, data-driven transportation decisions. StreetLight's proprietary data processing engine, Route Science®, algorithmically transforms its vast data resources to measure travel patterns of vehicles, bicycles, and pedestrians, accessible as analytics on the StreetLight InSight® SaaS platform. Acquired by Jacobs as a subsidiary in February 2022, StreetLight provides innovative digital solutions to help communities reduce congestion, improve safe and equitable transportation, and maximize the positive impact of infrastructure investment.



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